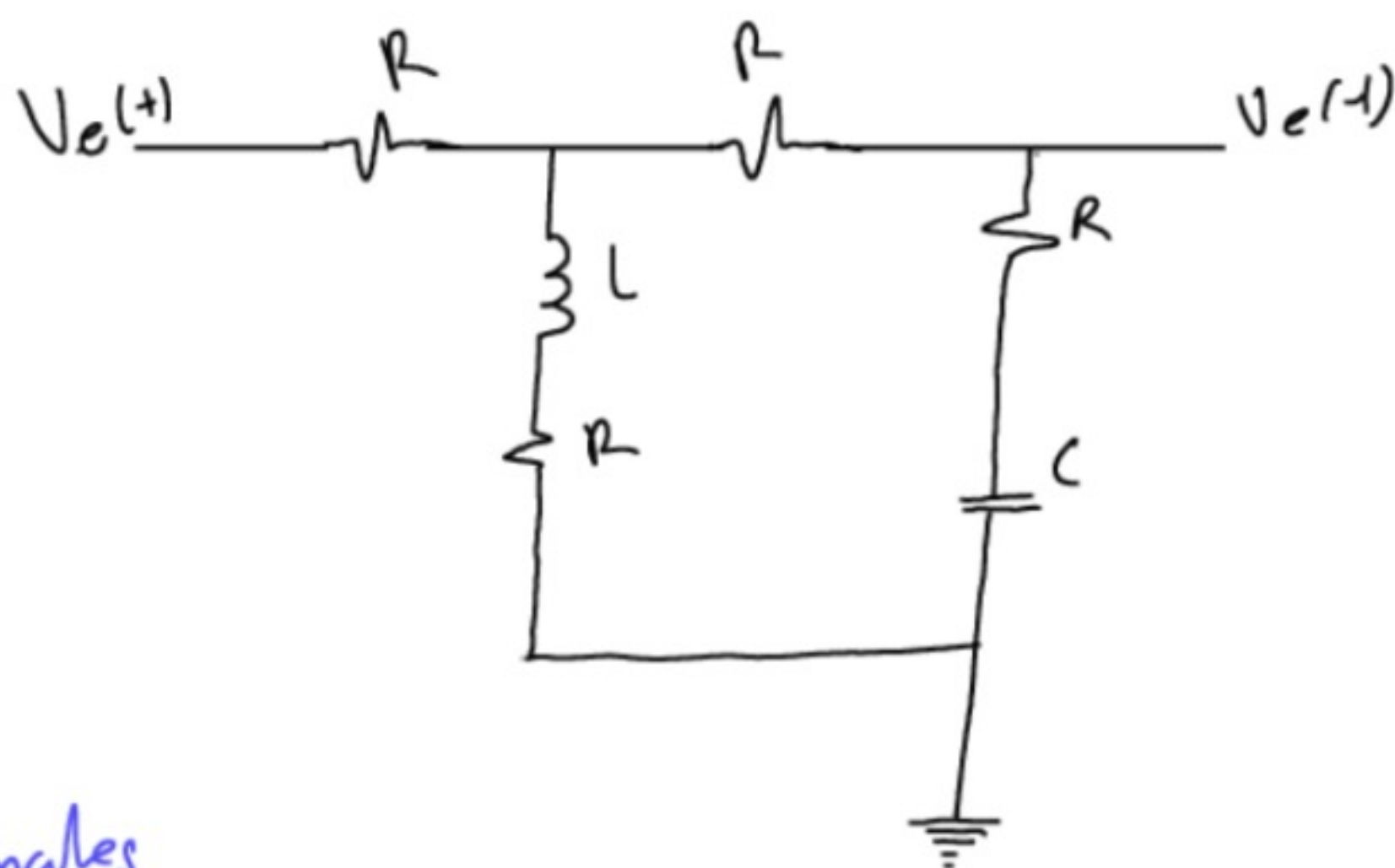


17/ Sep/ 2026



Ecuaciones Principales

$$V_e(t) = R \dot{i}_1(t) + L \frac{d(\dot{i}_1(t) - \dot{i}_2(t))}{dt} + R[\dot{i}_2(t) - \dot{i}_2(t)]$$

$$L \frac{d(\dot{i}_1(t) - \dot{i}_2(t))}{dt} + R[\dot{i}_2(t) - \dot{i}_2(t)] = R_2 \dot{i}_2(t) + R \dot{i}_2(t) + \frac{1}{C} \int \dot{i}_2(t) dt$$

$$V_s(t) = R \dot{i}_2(t) + \frac{1}{C} \int \dot{i}_2(t) dt$$

Transformadas de Laplace

$$V_e(s) = R \dot{i}_1(s) + Ls[\dot{i}_1(s) - \dot{i}_2(s)] + R(\dot{i}_1(s) - \dot{i}_2(s))$$

$$Ls[\dot{i}_1(s) - \dot{i}_2(s)] + R[\dot{i}_1(s) - \dot{i}_2(s)] = R \dot{i}_2(s) + R \dot{i}_2(s) + \frac{\dot{i}_2(s)}{Cs}$$

$$V_s(s) = R \dot{i}_2(s) + \frac{\dot{i}_2(s)}{Cs}$$

Procedimiento algebraico
- Función de Transferencia -

$$\frac{V_s(s)}{V_e(s)} = \frac{(\dots s^2 + \dots s + \dots) \dot{i}_2(s)}{\dots}$$

① Agrupar corrientes $\dot{i}_1(s)$ e $\dot{i}_2(s)$

② Expresar $V_e(s)$ en función de $\dot{i}_2(s)$

③ Despejar $\dot{i}_1(s)$ de la segunda ecuación

$$V_s(s) = \left(R + \frac{1}{Cs}\right) \dot{i}_2(s) = \frac{CRs + 1}{Cs} (\dot{i}_2(s))$$

$$L s \dot{i}_1(s) - L s \dot{i}_2(s) + R \dot{i}_1(s) - R \dot{i}_2(s) = R \dot{i}_2(s) + R \dot{i}_2(s) + \frac{\dot{i}_2(s)}{C s}$$

Simplifiquemos

$$(L s + R) \dot{i}_1(s) = (L s + \frac{1}{C s} + 3 R) \dot{i}_2(s)$$

$$\dot{i}_1(s) = \frac{1}{L s - R} (L s + \frac{1}{C s} - 3 R) \dot{i}_2(s)$$

$$\dot{i}_1(s) = \frac{1}{L s - R} \left[\frac{L C s^2 - 3 R C s + 1}{C s} \right] \dot{i}_2(s)$$

$$\dot{i}_1(s) = \frac{L (s^2 - 3 R C s + 1)}{C s (L s - R)} \dot{i}_2(s)$$

$$\dot{i}_1(s) = \left[\frac{L (s^2 - 3 R C s + 1)}{C s (L s - R)} \right] \dot{i}_2(s)$$

$$V_e(s) = R \dot{i}_1(s) + L s \dot{i}_1(s) - L s \dot{i}_2(s) + R \dot{i}_1(s) - R \dot{i}_2(s)$$

$$V_e(s) = \left[L s + 2 R \right] \frac{L C s^2 + 3 C R s + 1}{C L s^2 + C R s} \dot{i}_2(s) - (L s + R) \dot{i}_2(s)$$

$$V_e(s) = \left[\frac{(L s + 2 R) (C L s^2 + 3 C R s + 1) - (L s + R) (C L s^2 + C R s)}{C L s^2 + C R s} \right] \dot{i}_2(s)$$

$$V_e(s) = \left[\frac{C L^2 s^3 + 5 C L R s^2 + (C L + 6 C R^2) s + 2 R}{C L^2 s^3 + 2 C L R^2 + C R^2 s} \right] \dot{i}_2(s)$$

$$V_e(s) = \left[\frac{3 C L R s^2 + 5 C L R^2 s + 2 R}{C L s^2 + C R s} \right] \dot{i}_2(s)$$

Funcion de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{C R s + 1}{C s} (\dot{i}_2(s))}{\left[\frac{3 C L R s^2 + 5 C L R^2 s + 2 R}{C L s^2 + C R s} \right] \dot{i}_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(C R s + 1) (L s + R)}{3 C L R s^2 + (L + 5 C R^2) s + 2 R} = \frac{C L R s^2 + (C R^2 + L) s + R}{3 C L R s^2 + (L + 5 C R^2) s + 2 R}$$

Desarrollando FT

$$\frac{(R_s + 1)}{C_s} \left[\frac{C_s (Ls + R)}{3CLRs^2 + (L + S(R^2))s + 2R} \right]$$

$$\frac{V_s(s)}{V_c(s)} = \frac{(R_s + 1)(Ls + R)}{3CLRs^2 + (L + S(R^2))s + 2R}$$

$$L = 500 \mu H$$

$$C = 470 \mu F$$

$$R = 10 K \Omega$$

Error estacionario

$$e(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(1 - \frac{CLRs^2 + (CR + L)s + R}{3CLRs^2 + (L + S(R^2))s + 2R} \right)$$

$$\underline{e(t) = 0.5V}$$

Estabilidad en lazo abierto

$$3(470 \times 10^{-6})(500 \times 10^{-6})(10,000)s^2 + (550 \times 10^{-6} + 5(470 \times 10^{-6}))s + 20$$

Raices

$$x_1 = -0.085106$$

$$x_2 = -3.6303 \times 10^7$$

$$x_2 = -0.000006303$$

Respuesta: Sobreamortiguada

